VISHNU MUTHYALU

713-504-7131 | vm17college@gmail.com | Houston, TX | linkedin.com/in/vishnu-muthyalu/ | github.com/vishnumuthyalu

EDUCATION

The University of Texas at San Antonio, San Antonio, TX May 2025

Bachelor of Science, Computer Science, Cum Laude

- GPA : 3.55
- Relevant Coursework: Programming I/II, Data Structures, Application Programming, Systems Programming, Analysis of Algorithms, Database Systems, Programming Languages, Computer Architecture, Operating Systems, Artificial Intelligence, Machine Learning, Data Science, User Interfaces, Web Technologies, Deep Learning, and Software Engineering.

EXPERIENCE

REQUIRON ENGINEERING AND MARINE LLC., Houston, TX July 2025 - Present

Data Analyst

- Designed and implemented Windows Presentation Foundation (WPF) application tools to automate CSV dataset merging and streamline data preparation for oscillograph analysis platforms such as Yokogawa waveform software.
- Evaluated and validated Zigbee network devices by developing Python-based data acquisition scripts using MQTT libraries, enabling real-time data pull, storage, and preliminary integration of Zigbee sensor networks into marine energy monitoring systems.
- Engineered and tested secure VPN configurations across multiple commercial routers to optimize cross-site connectivity and ensure reliable remote access for systems across vessels.
- Programmed Siemens Programmable Logic Controllers (PLCs) in Structured Control Language (SCL) using TIA Portal, contributing to automation logic development and system-level process control.
- Utilized Siemens IBA PDA client tools to analyze high-frequency industrial data streams, generating actionable insights that supported system diagnostics and performance optimization.
- Developed multi-database connectivity with PostgreSQL and real-time visualization of data across multiple marine vessels for a web-based dashboard made with Node.js and React.js.
- Attended 2025 Marine Technology Society : Dynamic Position Conference in Houston from October 13-15th. Gathered insights on the importance of data centricity within marine risk safety and dynamic positioning.

CTV LAB | UT SAN ANTONIO, San Antonio, TX February 2025 - May 2025

Undergraduate Research Fellow

- Part of Dr. Guillermo Araya's Computational Turbulence and Visualization Lab at the University of Texas at San Antonio.
- Assisted in the development and implementation of GLTF (GL Transmission Format) asset integration within Unreal Engine, supporting optimization of 3D model loading and real-time rendering.
- Gained hands-on experience with flow-based visualization techniques using C++ in Unreal Engine to create dynamic user experiences.
- Learned to configure warehouse-style environments, including material setup for podiums, interactive flows, signs, and spatial organization within Unreal Engine.
- Learned and supported in creating effective 4K cinematics of flows using cinematic camera materials in Unreal Engine.

DEPARTMENT OF COMPUTER SCIENCE | UT SAN ANTONIO, San Antonio, TX January 2025 - May 2025
Computer Science Grader

- Graded assignments for the undergraduate Computer Architecture course at UT San Antonio (CS 3853) using a custom grading software (designed by Professor John Ortiz), ensuring accuracy and consistency in evaluations.
- Provided detailed and constructive feedback to students to enhance their understanding of topics like CPU architecture, memory hierarchy, instruction set design, as examples within the scope of Computer Architecture.
- Maintained organized records of grades and collaborated with the professor to address grading queries and improve the evaluation process.
- Deepened expertise in Computer Architecture concepts by analyzing and assessing student submissions on advanced topics.

UNIVERSITY TECHNOLOGY SOLUTIONS | UT SAN ANTONIO, San Antonio, TX January 2025 - May 2025
Bold Careers Program: ServiceNow Student Intern

- Gained hands-on experience with the ServiceNow platform in the Application Developer path.
- Learned to analyze data through Artificial Intelligence (AI) and Machine Learning (ML) for automation in ServiceNow.
- Learned to design, build, develop, and deploy applications in ServiceNow.
- Leveraged proficiency in JavaScript.
- Understood how ServiceNow can advance career opportunities.

NAJIM CENTER | UT SAN ANTONIO, San Antonio, TX September 2024 - November 2025
Strategist Program: Student Strategist at Community First Health Plans

- Collaborated with a team to develop strategies for Community First Health Plans, encouraging the first-line use of psychosocial care over mental illness medications for primary care providers (PCPs).
- Conceived and demonstrated the proposal of a chatbot integrated into the Community First website to improve navigation and accessibility for both PCPs and plan members.
- Recommended AI-Driven Personalization to tailor resources for PCPs, enhancing their awareness and encouraging the adoption of psychosocial care.
- Advocated for AI-Generated Reports to provide actionable insights for PCPs, enabling them to track their utilization of psychosocial care and identify areas for improvement.
- Contributed to bridging the gap between technology and healthcare by promoting innovative AI solutions to enhance psychosocial care compliance.

PARINT LAB | UT SAN ANTONIO, San Antonio, TX September 2024 - January 2025
Undergraduate Research Fellow

- Part of Dr. Sushil Prasad's ParInt Lab at UT San Antonio.
- Collaborated closely with a PhD candidate, investigated and conducted research on polygon clipping and polygon overlay techniques.
- Gained proficiency in GPU programming, utilizing CUDA with C and C++.
- Developed strong research skills by exploring advanced computational geometry concepts and contributing to discussions on algorithm design and implementation.
- Contributed to the development and optimization of algorithms for efficient polygon operations, leveraging parallel computing for increased performance.

ASPIN KEMP AND ASSOCIATES INC. (AKA), Prince Edward Island, Canada May 2024 - June 2024
Computer Science Student Intern

- Gained knowledge of source code creation (in JavaScript) for AKA-designed software used in controlling and analyzing energy systems, as well as coding for Programmable Logic Controllers (in Structured Command Language), and other AKA designed-components (in C).
- Developed strong organizational abilities, critical thinking skills, and effective listening and questioning skills throughout the internship.
- Set up, managed, and troubleshooted Windows Virtual Machines on Linux OS; reviewed schematics and assembled PCs for data logging.
- Used IBA Analyzer software for data analysis by creating templates of battery data signals coming from AKA designed power conversion systems.
- Internship done under the supervision of Jason Aspin (CEO/CTO), Ed Bourgeau (Chief Engineer), and Christian Wiedemer (Head of Data Analytics).

NAJIM CENTER | UT SAN ANTONIO, San Antonio, TX February 2024 - April 2024

Strategist Program: Student Strategist at Inov8: office and event spaces

- Collaborated closely with Mrs. Yvonne "Bonnie" Garcia, CEO of Inov8 - meeting and event spaces.
- Led comprehensive market research initiatives to evaluate pricing strategies and formulate pricing models for Inov8.
- Leveraged Python programming for proficient data analysis and visualization.
- Strengthened competencies in teamwork, leadership, critical thinking, and communication.

COLLEGE OF SCIENCES | UT SAN ANTONIO, San Antonio, TX August 2023 – December 2024

Undergraduate Research Fellow

- Served as a tutor within the College of Sciences, providing both drop-in and scheduled tutoring sessions for students.
- Specialized in assisting with math and computer science courses offered at UT San Antonio.
- Maintained meticulous records for each tutoring session, monitoring student advancement, and identifying specific areas in need of additional attention.
- Reinforced personal mastery of foundational computer science concepts such as data structures, algorithms, and computational logic through continuous problem-solving and peer explanation.
- Strengthened ability to articulate complex programming and mathematical topics clearly, enhancing both communication skills and conceptual depth in core CS principles.

PROJECTS

- **WORKLOGGER**
  - Built a full-stack productivity application using Python and Streamlit, featuring interactive calendar widgets, real-time data visualization, and multi-format document generation (CSV, Word, Text).
  - Designed and implemented a robust database layer with SQLAlchemy ORM (Object Relational Mapper) and SQLite backend, including automated schema management, data integrity constraints, and session state optimization.
  - Developed a comprehensive export and email integration system with secure SMTP functionality, SSL/TLS encryption, and professional document styling for seamless workflow automation.
  - Engineered modular code architecture with clean separation of concerns across database utilities, export functions, email services, and configuration management, demonstrating modern software development practices.
- **ELECTRONICS STORE**

- Architected a full-stack e-commerce platform using React.js with React Router DOM (Document Object Model) for single page application navigation, Firebase Authentication for secure user access, and Express.js RESTful API backend integrated with Stripe for real-time payment processing.
- Designed modular, reusable UI components following component-based design principles and implemented Context API for global state management (Cart, Auth, Search) with persistent storage via localStorage.
- Engineered backend utility modules for CSV-based inventory and discount-code management using csv-parse/PapaParse, enabling dynamic product updates and admin-level product control.
- Deployed the full-stack application on AWS EC2 (Amazon web services elastic compute cloud) , configuring CORS (Cross-Origin Resource Sharing) middleware, environment variables, and secure API access through HTTPS for cloud scalability and global availability.
- Integrated Stripe.js SDK and Firebase SDK for end-to-end checkout sessions, achieving a responsive, cross-platform interface optimized for desktop and mobile shopping experiences.

- **FACIAL RECOGNITION AND AGE/EMOTION DETECTION**
  - Collaborated on a real-time system integrating facial recognition, emotion detection, and age prediction using OpenCV and CNNs (Convolutional Neural Networks).
  - Led data preprocessing and model training for emotion detection using the FER-2013 dataset, achieving 72% categorical accuracy.
  - Implemented real-time emotion and age prediction with a processing speed of ~18 FPS on standard hardware.
  - Trained CNNs for emotion detection and age prediction over 50 epochs, achieving a MAE (Mean Absolute Error) of 3.8 years for age prediction.
  - Optimized data pipelines for efficient processing of live video feeds, ensuring reliable performance on standard hardware configurations.
- **GEARBOX SUPPLY – AUTO PARTS E-COMMERCE WEBSITE**
  - Developed a dynamic e-commerce website using React.js for selling automotive parts - Integrated Firebase to store inventory, car service information, and cart items with real-time database updates.
  - Designed and implemented a navigation bar menu that directs users to different pages such as:
    - Homepage: Displays featured auto parts and services (through random selection).
    - Shop Page: displays parts as categories, which can then be expanded to show just the parts for those categories.
    - Services Page: to view services available.
    - For Your Car Page: to add information precise to your car.
    - GearBot: Integrated a chatbot that utilizes Gemini Flash API.
    - Contact Us Page: displays team along with email buttons - Search box: to search up items in the inventory.
    - Account Page: Implemented user authentication using Firebase Authentication for secure login and sign-up.
    - Cart Page: add or delete items, update item quantities, view total checkout price.
  - Item are updated in real time on firebase.
  - Leveraged React.js to create a responsive and user-friendly interface, prioritizing usability and real-time updates to enhance the customer experience.

- **MYRUNBUDDY ANDROID APPLICATION**
    - Developed a full-stack mobile application in Android Studio utilizing Java and Kotlin, designed to cater to running enthusiasts. The app boasts a user-friendly interface with multiple screens, intuitive buttons, and interactive text boxes for seamless navigation.
    - Menu Options: The application includes dedicated menu options for logging detailed run metrics and accessing a custom-built chatbot, RunBuddy, for instant running-related advice.
    - Comprehensive Running Log: Leveraged SQLite for robust database management, enabling users to efficiently insert, update, delete, and view their running logs. Each log captures critical details such as run name, run date, run duration, distance covered, average pace, average BPM.
    - Chatbot Integration with Google's Gemini API: Integrated Google's Gemini API to develop a sophisticated chatbot using the Gemini-1.5-flash model. RunBuddy offers personalized guidance on running, health, and fitness, making it a valuable companion for runners of all levels.
    - User Experience: Emphasized a clean, intuitive design with responsive elements to deliver an engaging and smooth user experience across various Android devices.
- **CAR DEALERSHIP DATABASE**
    - Collaborated on a 3NF entity relationship diagram for a sophisticated car dealership system, encompassing various components such as customer data, vehicle inventory, suppliers, and more.
    - Developed MySQL tables covering customer information, vehicle inventory, service records, sales transactions, and price audits to ensure comprehensive data management.
    - Utilized MySQL expertise to craft efficient queries, establish real-time update views for service progress tracking, and engineered stored procedures for dynamic inventory cost calculation and price adjustments based on market trends, with automatic table updates triggered by price changes.
- **CORONAVIRUS IN TEXAS**
    - Implemented concepts of Python File Reading, Graph Creation, and Searching Algorithms.
    - Used Pandas libraries to read data from .csv files that contain the list of Texas Cities, and Distance between two cities (if there is a path present).
    - Used matplotlib.pyplot, networkkx, matplotlib.image to create the graph that represents the map of Texas and plotted it over an image of the map of the state of Texas.
    - Used concepts of nodes to represent the cities in the graphs.
    - Performed Breadth First Search Algorithm as a uniformed search agent that traverses through the graph (map of Texas) starting from "Three Rivers", In which it hits each city (node) at least once and records the total distance travelled. This was done to mimic the spread of the virus starting from "Three Rivers."
    - Used A* Algorithm as an informed search agent to find the optimal path from "San Antonio" (start node) to "College Station" (goal node). Implemented calculations of Euclidean Distance as a heuristic function. This was done to mimic the scenario of finding an optimal of transferring a vaccine from San Antonio (city where vaccine is in supply) to College Station (city, where vaccine is in demand).
- **MACHINE LEARNING MODEL TRAINING AND TESTING**

- Learnt to implement and develop machine learning models using Stochastic Gradient Descent (SGD) and Iterative Dichotomiser 3 (ID3) algorithms on specific datasets for training and testing.
- Applied Activation Function and Perceptron Algorithm concepts to implement the SGD algorithm efficiently.
- For SGD, steps involved data loading, model training with iterative weight adjustment, and accuracy evaluation on test datasets.
- Implemented the ID3 algorithm by creating functions for entropy calculation, information gain, maximum gain, tree structure, and recursive ID3.
- Utilized ML concepts such as supervised learning for binary classification, optimization techniques like gradient descent in SGD, and decision tree construction using entropy and information gain in ID3.

LEADERSHIP & COMMUNITY INVOLVEMENT

CORROSION LAB | UT SAN ANTONIO, San Antonio, TX July 2025 - October 2025

Full Stack Web Developer

- Developed a modern React.js website showcasing the UT San Antonio Corrosion Research Laboratory's academic research, equipment, and team using React 19, React Router, and Vite for optimal performance.
- Created comprehensive content management system featuring dynamic pages for research projects, publications, staff profiles, equipment galleries, and STEM outreach programs with responsive design
- Implemented interactive UI components including image carousels, publication timelines, staff directories with role-based filtering, and equipment showcases with detailed specifications and galleries.
- Built responsive academic portfolio features displaying 20+ research publications, 15+ specialized equipment pieces, graduate/undergraduate researcher profiles, and laboratory capabilities with professional styling.
- Deployed modern web technologies utilizing Lucide React icons, CSS Grid/Flexbox layouts, and component-based architecture to create an accessible academic website promoting corrosion science research.

GLOBAL CITIZENS' ASSOCIATION | UT SAN ANTONIO, San Antonio, TX January 2025 - May 2025

Social Media Marketing Specialist and Photographer

- Designing flyers, social media posts, and promotional materials for organizational events, fundraisers, and activities, ensuring clear and engaging messaging.
- Managing and updating social media platforms and links, and maintaining a consistent and professional online presence.
- Capturing events for use in social media posts, promotional content, and archival purposes.

TECHNICAL SKILLS

Programming Languages: Proficient in Python, C/C++, Java, SQL, C#, Kotlin, JavaScript, R, HTML/CSS, MATLAB, and Bash/Shell.

Frameworks & Libraries: TensorFlow, PyTorch, scikit-learn, NumPy, Pandas, Matplotlib, OpenCV, React.js, Node.js, .NET, ASP.NET MVC, Unreal Engine 5, Streamlit, MySQL, PostgreSQL, Google Firebase, and SQLite.

Tools & Platforms: Git, Docker, Kubernetes, GitHub Actions, Jupyter Notebook, Anaconda, Visual Studio, Visual Studio Code, Linux, Windows, and AWS.

Languages: English, Hindi, and Telugu.

Certifications:

- IBM Full-Stack JavaScript Developer – Professional Certificate
- IBM Introduction to Data Science – Specialization Certificate

HONORS AND AWARDS

- **President's List** : Fall 2024 and Spring 2025
- **Dean's List** : Fall 2022, Fall 2023, and Spring 2024